Sleep Health and Lifestyle Analysis

1. Project Overview

This project analyzes lifestyle and sleep data of 374 individuals across various occupations. The goal is to uncover insights into sleeping patterns, physical activity patterns and identify relationships between stress, physical activity, BMI, and sleep quality using Python, SQL, and Power BI to guide people lead a healthy life.

2. Dataset Summary

- Rows: 374 – Columns: 13 – Key Features:
- Person Details (Age, Gender, Occupation, BMI Category)
- Lifestyle Behavior (Sleep Duration, Quality of Sleep, Physical Activity Level, Stress Level, Daily steps, Sleep Disorder)
- Medical Parameters (Blood Pressure, Heart Rate)
- No Missing values
- No Duplicate values

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used df.info() to check structure and .describe() for summary statistics.

```
        Person_ID         Age  Sleep_Duration  Quality_of_Sleep  \
count  374.000000  374.000000      374.000000        374.000000
mean   187.500000   42.184492        7.132086          7.312834
std    108.108742    8.673133        0.795657          1.196956
min      1.000000   27.000000        5.800000          4.000000
25%     94.250000   35.250000        6.400000          6.000000
50%    187.500000   43.000000        7.200000          7.000000
75%    280.750000   50.000000        7.800000          8.000000
max    374.000000   59.000000        8.500000          9.000000

       Physical_Activity_Level  Stress_Level  Heart_Rate   Daily_Steps
count               374.000000    374.000000  374.000000    374.000000
mean                 59.171123      5.385027   70.165775   6816.844920
std                  20.830804      1.774526    4.135676   1617.915679
min                  30.000000      3.000000   65.000000   3000.000000
25%                  45.000000      4.000000   68.000000   5600.000000
50%                  60.000000      5.000000   70.000000   7000.000000
75%                  75.000000      7.000000   72.000000   8000.000000
max                  90.000000      8.000000   86.000000  10000.000000
```

- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Database Integration:** Connected Python script to MYSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL

We performed structured analysis in MYSQL to answer key questions:

1. Top 3 occupations with highest average stress level

avgstress	Occupation
4.5250	Teacher
6.0000	Software Engineer
7.0000	Scientist

2. Top 5 most active people (Daily Steps)

Person_ID	Gender	Age	Occupation	Sleep_Duration	Quality_of_Sleep	Physical_Activity_Level	Stress_Level	BMI_Category	Blood_Pressure	Heart_Rate	Daily_Steps	Sleep_Disorder
306	Female	51	Nurse	6.1	6	90	8	Overweight	140/95	75	10000	Sleep Apnea
2	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	None
3	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	None
305	Female	51	Nurse	6.1	6	90	8	Overweight	140/95	75	10000	Sleep Apnea
291	Female	50	Nurse	6	6	90	8	Overweight	140/95	75	10000	Sleep Apnea

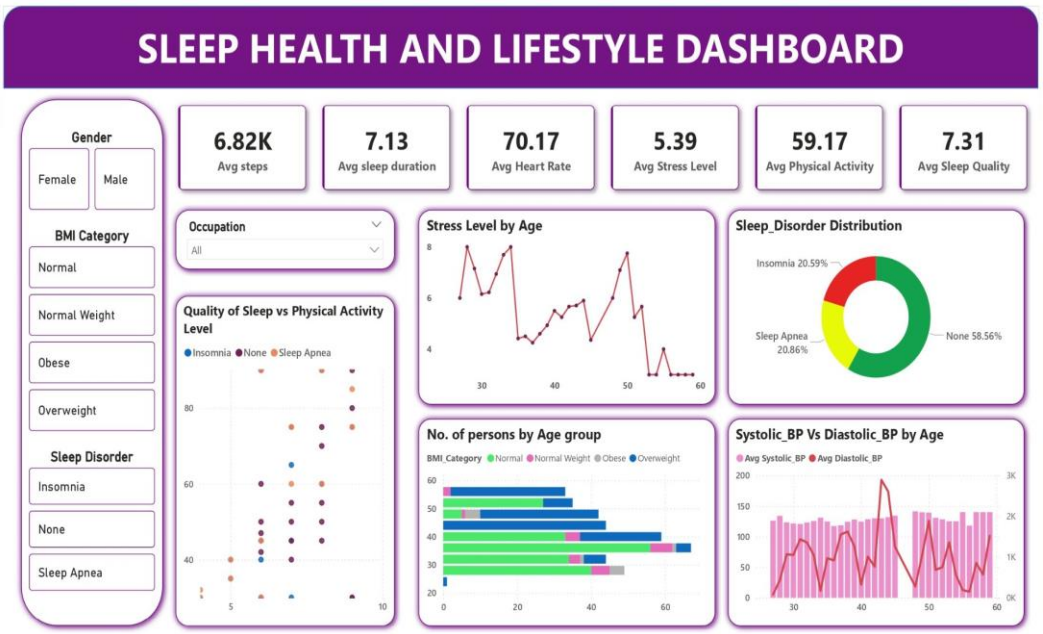
3. BMI Category VS Highest stress level

BMI_Category	avg_stress
Overweight	5.7297
Obese	5.7000
Normal Weight	5.1905
Normal	5.1282

5. Dashboard (Power BI)

Interactive dashboard with KPIs and visual insights:

Metric	Value
Avg Steps	6.82K
Sleep Duration	7.13 hrs
Stress	5.39
Sleep Quality	7.31



6. Conclusion

Lifestyle factors such as physical activity, stress, and BMI significantly impact sleep health. The dashboard provides actionable insights for improving well-being.

- People with lower stress levels have sleep duration of more than 7 hours.
- Females have more average sleep duration as compared to males.
- Males have comparatively higher average stress levels than females.
- Nurses have the highest no. of daily steps whereas Sales Representatives have the lowest.
- People with No Sleep Disorder have sleep duration of more than 7 hours , while people with Insomnia have least sleep duration i.e less than 7 hours.
- Overweight people have higher stress levels whereas normal people have less stress levels.